

Algebra II - Tutorial 1 - Date: 12/1/2026

1. Let x, y be elements of a group G . Assume that x, y, xy are elements of order 2. Show that $H = \{1, x, y, xy\}$ is a subgroup of order 4.
2. A permutation that can be expressed as a even product of 2-cycles is called an even permutation. Let A_n denote the set of all even permutations. Show that A_n is a subgroup of S_n .
3. List all subgroups of the dihedral group D_4 . Which of them are normal. Is it possible to list all subgroups of D_n ($n \geq 3$)
4. List all subgroups of the symmetric group S_4 . Which of them are normal?
5. Let $H = \{1, x^5\}$ be a subgroup of the dihedral group $G = D_{10} := \langle x, y : x^{10} = y^2 = 1, yx = x^{-1}y \rangle$. Compute all the left cosets G/H .
6. A n -th root of unity is a complex number z such that $z^n = 1$. (A cyclic group is a group generated by one element).
 - (a) Show that the n -th roots of unity form a cyclic subgroup of $\mathbb{C}^*$.
 - (b) Determine the product of all n -th roots of unity.
7. Show that the elementary matrices (Page 71 of Artin)
of first type, $E_1 = \left(\text{identity with } a \text{ at position } (j, i)\right)$ and
third type, $E_3 = \left(\text{identity with } c \text{ at position } (i, i)\right)$ generate $GL_n(\mathbb{R})$.
8. Show that elementary matrices of first type generate $SL_n(\mathbb{R}) := \{A \in GL_n(\mathbb{R}) : \det(A) = 1\}$.
9. (a) Let $G = S_n$. Use the row reduction method to show that transpositions generate the symmetric group S_n .
(b) Show that for $n \geq 3$, three cycles generate the alternating group A_n .

10. Prove that the set of all $n \times n$ matrices that have block form $M = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$ with $A \in GL_r(\mathbb{R})$ and $D \in GL_{n-r}(\mathbb{R})$ form a subgroup say H of $GL_n(\mathbb{R})$ and that the map that sends $\phi : H \rightarrow GL_r(\mathbb{R})$ that sends $M \in H$ to $A \in GL_r(R)$ is a homomorphism. What is its kernel? ($\ker(\phi) := \{M \in H : \phi(M) = I_r\}$).
11. Let $G = GL_n(\mathbb{R})$ be the group of all $n \times n$ invertible matrices. $Z(G) := \{g \in G \mid xg = gx \text{ for all } x \in G\}$.
12. Consider the elementary matrices $E_1 = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$ and $E_2 = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$ where $a, b \in \mathbb{R}$. Show that $SL_2(\mathbb{R})$ can be generated by these elementary matrices. (Hint: Perform row / column reduction on matrices in $SL_2(\mathbb{R})$)
13. Let G be a group. An automorphism is an isomorphism $\phi : G \rightarrow G$. Let $Aut(G) : \{\phi : G \rightarrow G \mid \phi \text{ is an isomorphism}\}$. Show that $Aut(G)$ is a group. Determine $Aut(\mathbb{Z}_n)$.
14. Show that in an abelian group every subgroup is normal. Let $G = Q_8$, the quaternion group. List all subgroups of Q_8 . Which of them are normal.